

HEMANTH VASUDEV N P

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EDUCATION

PSG College of Technology
M.Sc (Integrated) Software Systems

Present
CGPA: 8.43 / 10

The TVS School, Madurai
High School

June 2024
Percentage: 90.3

SKILLS

Languages: Python, C++, C, JavaScript
Tools: Git, Redis, Prometheus, Postman
Databases: MySQL, PostgreSQL, MongoDB
Frameworks: FastAPI, Flask, ReactJS, Node.js, scikit-learn, NumPy, SciPy, sentence-transformers
DevOps & Cloud: Docker, Kubernetes, Linux, GCP, Azure

PROJECTS

CineScope — Film Discovery & Recommendation Platform |

- Hybrid recommender (TF-IDF + TruncatedSVD, 20 latent factors) with a sparsity-aware 75/25 content-collaborative split; re-ranking via preference-bucket scoring and a quality formula with relaxed vote thresholds for non-English cinema.
- NL query pipeline: rule-based intent classification (8 types), SBERT semantic search (all-MiniLM-L6-v2) with TF-IDF fallback, asyncio parallel tool execution, and per-service circuit breakers.
- Offline evaluation suite — Hit Rate@K, Precision@K, MRR, NDCG@K via leave-one-out; 71 pytest tests gate a GitHub Actions pipeline building and deploying 5 Dockerized services.

Throttlr — High-Performance API Gateway |

- C++20 epoll gateway on Linux — 50K+ req/s, ~300 μ s P50 / ~1.8 ms P99 (wrk, 4-core/16 GB); JWT HS256 auth, token bucket rate limiting backed by Redis atomic Lua scripts.
- 3-state circuit breaker FSM (count + failure-rate trip), four load balancing strategies including consistent hash (150 virtual nodes), and a LIFO connection pool with timerfd idle eviction.
- Prometheus metrics, structured JSON logs, Kubernetes rolling-update manifests (3 replicas, liveness/readiness probes), and SIGHUP hot reload with zero dropped connections.

MargaMetis — Intelligent Route Optimizer |

- Dijkstra, A*, Bidirectional A*, and Yen's K-Shortest from scratch on a Chennai OSM graph (22K nodes, 55K edges); BiDir-A* hit 6 ms vs Dijkstra's 60 ms, exploring 12.9 $\times$ fewer nodes (1,470 vs 18,954).
- Groq LLaMA 3.1 with few-shot prompting converts natural-language route preferences into structured constraint JSON; cost weights injected into the graph traversal at runtime, with rule-based fallback.
- Redis caching (route 1h, geocode 24h) cut repeat query latency from 3,500 ms to 16 ms; deployed as a 4-service Docker Compose stack on Vercel and Railway.

ACTIVITIES & HOBBIES

CSA Tech Team Tech Team member of the Computational Sciences Association (CSA), PSGCT

FinVerse Analyst on PSGCT's FinVerse finance club

Linux & OSS Explore Linux distros and contribute to open-source projects

Photography Amateur photography with a focus on composition and detail